

GATE VALVE MAINTENANCE MANUAL — VALVE-V12

Equipment Tag: Valve-V12 | Type: Motor-Operated Gate Valve | Rev. 2 (2021)

PipeFlow Engineering Corp | Document No: PFE-MNT-GV-012 | CONTROLLED COPY

SECTION 1 — EQUIPMENT IDENTIFICATION

This document provides maintenance, inspection, and operational guidance for Valve-V12, a DN150 motor-operated gate valve installed on the high-pressure process gas supply line in Process Area B. Valve-V12 is a safety-critical isolation device. Deviation from this maintenance schedule must be approved by the Plant Safety Engineer in writing.

Equipment Tag: Valve-V12 | Type: Motor-Operated Gate Valve | Size: DN150 (6 inch) | Pressure Rating: ANSI Class 300 (740 psi CWP) | Body Material: Carbon Steel ASTM A216 WCB | Actuation: Electric motor actuator, 480V 3-phase | Installed: 2018-06-20 | Serial No: PFE-GV-2018-0512

SECTION 2 — TECHNICAL SPECIFICATIONS

Maximum Allowable Working Pressure (MAWP): 740 psi at 100 °F

Design Temperature Range: -20 °F to +450 °F

Seat Leakage Class: ANSI/FCI 70-2 Class IV (0.01% of rated capacity)

Stem Diameter: 50 mm | Stem Material: AISI 316 Stainless Steel

Actuator Torque Output: 900 N·m | Actuator Duty: S2-15 min

Open/Close Travel Time: 45 seconds at rated torque

Packing Type: Graphite chevron rings (5-ring set), PN-GR-150

SECTION 3 — INSPECTION SCHEDULE

[WARNING] Valve-V12 is a SAFETY-CRITICAL isolation valve on a high-pressure process line. The inspection interval MUST NOT be extended without a formal risk assessment signed by the Plant Safety Engineer and the Operations Manager.

3.1 Required Inspection Frequency

Valve-V12 shall be inspected every 3 months (90 days). This interval is mandated by the OEM based on the valve's service conditions (high-pressure cycling, elevated temperature, potential for packing degradation). Failure to comply with this schedule voids the OEM warranty and constitutes a documented safety deviation that must be reported to the HSE Department.

3.2 Quarterly Inspection Checklist (every 90 days)

- Perform open/close cycle test — verify full travel within 45 ± 5 seconds.

- Inspect stem packing for weeping or visible leakage; replace if any leakage detected.
- Torque packing gland nuts to 35 N·m ± 5 N·m (do not over-torque).
- Lubricate stem threads with approved thread compound (Molykote P-37 or equivalent).
- Verify actuator limit switches — confirm open and closed position indication.
- Check actuator motor insulation resistance (minimum 1 MΩ at 500V DC).
- Inspect body for external corrosion, coating damage, and mechanical damage.
- Record all findings in Work Order and Valve Inspection Report Form EQ-VI-012.

3.3 Annual Inspection (in addition to quarterly)

- Full disassembly and internal inspection of gate, seat rings, and body cavity.
- Replace packing set and seat rings regardless of apparent condition.
- Actuator gearbox oil change (Shell Omala S2 G 220 or equivalent).
- Hydrostatic shell test at 1,110 psi ($1.5 \times \text{MAWP}$) for 10 minutes.

SECTION 4 — MAINTENANCE PROCEDURES

4.1 Packing Replacement Procedure

1. Isolate valve from process — confirm upstream and downstream blind flanges are installed.
2. De-energise actuator at local disconnect; lock out and tag out (LOTO Procedure SAFE-LOTO-007).
3. Remove packing gland follower and extract worn packing rings.
4. Clean stem and stuffing box bore with lint-free cloth; inspect for scoring.
5. Install new graphite rings (PN-GR-150) — stagger cuts by 90° minimum.
6. Replace gland follower and torque gland nuts to 25 N·m (initial).
7. Re-pressurise and perform packing leak test; tighten to 35 N·m if weeping persists.
8. Restore power and perform full open/close cycle test. Document in work order.